Monte Carlo Simulation of Neutron Transport

First-Principles Verification of Exponential Attenuation in Nuclear Materials

Stage II of a Progressive Monte Carlo Research Project in Nuclear Physics

Auhornish Adel

April 2026

Abstract

This study presents a Monte Carlo simulation of neutron transport through homogeneous material slabs, developed as a direct progression from Stage I Monte Carlo estimation of π. By sampling free path lengths from the exponential distribution $f(d) = \Sigma e^{-\Sigma d}$ via inverse transform sampling, the simulation recovers the Beer-Lambert attenuation law $P(x) = e^{-\Sigma x}$ without assuming its functional form. Results across a range of slab thicknesses and four reactor-relevant materials demonstrate agreement between Monte Carlo estimates and analytical predictions to within the theoretical statistical uncertainty $\sigma_T = \sqrt{\hat{T}(1-\hat{T})/N}$. The $1/\sqrt{N}$ convergence behaviour observed in Stage I reappears identically, confirming the universality of this property across Monte Carlo methods. These findings establish the conceptual and computational foundation for Stage III full reactor simulation using OpenMC.

Contents

1 Introduction **3**

2 Theoretical Framework **3**

2.1 The Macroscopic Cross-Section 3

2.2 Derivation of Exponential Attenuation 4

2.3 The Mean Free Path 5

2.4 Free Path Sampling via Inverse Transform 5

3 Statistical Framework **6**

3.1 The Monte Carlo Estimator 6

3.2 Uncertainty Quantification 6

4 Simulation Methodology **7**

4.1 Implementation 7

4.2 Experimental Design 7

5 Results **8**

5.1 Transmission vs Thickness 8

5.2 Statistical Convergence 9

5.3 Material Comparison 10

6 Discussion **10**

7 Computational Analysis **11**

8 Progression to Stage III: Full Reactor Simulation **12**

9 Conclusion **12**

1 Introduction

Monte Carlo methods are computational techniques that rely on random sampling to solve problems that are probabilistic in nature. In nuclear engineering, they form the basis of production reactor analysis codes such as OpenMC, MCNP, and Serpent, which simulate billions of individual neutron histories to compute reactor criticality, neutron flux distributions, and radiation shielding characteristics. Each of these codes is built on the same fundamental principle implemented in this study.

In Stage I of this research project, Monte Carlo methods were applied to estimate π through random geometric sampling. By generating uniformly distributed points and evaluating their position relative to a quarter circle, an approximation of π was obtained. The key findings were that accuracy improves with sample size following a $1/\sqrt{N}$ convergence law, and that computational cost scales linearly with N. Stage II applies the same probabilistic framework to a physically meaningful problem: the transmission of neutrons through a material slab.

Neutrons are uniquely suited to Monte Carlo simulation. Unlike charged particles, which decelerate continuously through electromagnetic interactions with atomic electrons, neutrons carry no electric charge and travel in straight lines between discrete collision events. Each collision is memoryless: the probability of a collision in the next infinitesimal distance dx is constant, regardless of how far the neutron has already traveled. This property determines the statistical distribution of free path lengths and leads directly to exponential attenuation.

This paper derives the exponential attenuation law from the collision rate postulate, implements the simulation using inverse transform sampling, and validates results against the analytical Beer-Lambert law across a range of thicknesses and materials. The convergence behaviour is characterised and connected to the findings of Stage I. Stage III will extend this single-collision model to full multi-collision transport using OpenMC, incorporating scattering, energy dependence, and three-dimensional reactor geometry.

2 Theoretical Framework

2.1 The Macroscopic Cross-Section

The macroscopic cross-section Σ characterises the collision density of a material:

$$\Sigma = n \cdot \sigma \quad (\mathrm{cm}^{-1}) \tag{1}$$

where n is the atomic number density (atoms/cm^3) and σ is the microscopic cross-section (cm^2). The microscopic cross-section is not the physical size of the nucleus but a quantum mechanical interaction probability expressed as an equivalent area. For thermal neutrons,

σ can exceed the geometric nuclear cross-section by orders of magnitude due to resonance effects. Table 1 lists the values used in this study.

Material	Σ **(cm^{-1})**	$\lambda = 1/\Sigma$ **(cm)**
Light water	3.00	0.33
Iron	1.60	0.63
Graphite	0.38	2.63
Heavy water	0.18	5.56

Table 1: Macroscopic cross-sections and mean free paths for thermal neutrons in selected reactor-relevant materials.

2.2 Derivation of Exponential Attenuation

Let $P(x)$ denote the probability that a neutron survives to depth x without any collision. The collision rate postulate states that the probability of a collision occurring in an infinitesimal layer of thickness dx, given survival to x, is:

$$P(\text{collision in } dx \mid \text{survived to } x) = \Sigma\, dx \tag{2}$$

This probability is independent of x, reflecting the memoryless nature of nuclear collisions. The survival probability at depth $x + dx$ is therefore:

$$P(x + dx) = P(x)(1 - \Sigma\, dx) \tag{3}$$

Expanding and rearranging:

$$P(x + dx) - P(x) = -\Sigma \cdot P(x) \cdot dx \tag{4}$$

Dividing both sides by dx and taking the limit as $dx \to 0$ yields the first-order ordinary differential equation:

$$\frac{dP}{dx} = -\Sigma \cdot P(x) \tag{5}$$

Equation (5) is solved by separation of variables. Dividing both sides by P and multiplying by dx:

$$\frac{dP}{P} = -\Sigma\, dx \tag{6}$$

Integrating both sides:

$$\int \frac{dP}{P} = \int -\Sigma\, dx \quad \Longrightarrow \quad \ln P = -\Sigma x + C \tag{7}$$

Exponentiating:

$$P(x) = Ae^{-\Sigma x} \tag{8}$$

Applying the boundary condition $P(0) = 1$ gives $A = 1$, and therefore:

$$\boxed{P(x) = e^{-\Sigma x}} \tag{9}$$

Equation (9) is the Beer-Lambert attenuation law. It was not assumed but derived from the single postulate of a memoryless, constant collision rate. The simulation constitutes a numerical verification of this derivation: exponential attenuation emerges from random sampling alone, without any assumed functional form.

2.3 The Mean Free Path

The mean free path λ is the average distance a neutron travels between collisions. The probability density function of free path lengths is:

$$f(d) = \Sigma e^{-\Sigma d}, \quad d \geq 0 \tag{10}$$

The mean is computed by integration by parts, with $u = d$ and $dv = \Sigma e^{-\Sigma d}\, dd$:

$$\langle d \rangle = \int_0^\infty d \cdot \Sigma e^{-\Sigma d}\, dd = \Big[- d e^{-\Sigma d} \Big]_0^\infty + \int_0^\infty e^{-\Sigma d}\, dd = 0 + \frac{1}{\Sigma} \tag{11}$$

Therefore:

$$\lambda = \frac{1}{\Sigma} \tag{12}$$

2.4 Free Path Sampling via Inverse Transform

To sample free path lengths from $f(d) = \Sigma e^{-\Sigma d}$ using a uniform random number $r \in (0, 1)$, the cumulative distribution function is first derived:

$$F(d) = \int_0^d \Sigma e^{-\Sigma t}\, dt = 1 - e^{-\Sigma d} \tag{13}$$

Setting $F(d) = r$ and solving for d:

$$r = 1 - e^{-\Sigma d} \tag{14}$$

$$e^{-\Sigma d} = 1 - r \tag{15}$$

$$d = -\frac{\ln(1 - r)}{\Sigma} \tag{16}$$

Since $(1-r)$ is also uniformly distributed on $(0,1)$, this simplifies to:

$$\boxed{d = -\frac{\ln r}{\Sigma}} \tag{17}$$

Equation (17) is exact. Every sampled value of d follows the true physical distribution $f(d)$ with no approximation or discretisation.

3 Statistical Framework

3.1 The Monte Carlo Estimator

Each neutron history constitutes an independent Bernoulli trial. A neutron either transmits through the slab (outcome = 1) with probability $p = e^{-\Sigma x}$, or is absorbed (outcome = 0) with probability $1-p$. The Monte Carlo estimator of the transmission fraction is:

$$\hat{T} = \frac{N_{\text{trans}}}{N} \tag{18}$$

By the Law of Large Numbers, $\hat{T} \to p$ as $N \to \infty$.

3.2 Uncertainty Quantification

For a single Bernoulli trial X:

$$\text{Var}(X) = E[X^2] - (E[X])^2 = p - p^2 = p(1-p) \tag{19}$$

The variance of the mean of N independent trials is:

$$\text{Var}(\hat{T}) = \frac{p(1-p)}{N} \tag{20}$$

The standard uncertainty is therefore:

$$\sigma_T = \sqrt{\frac{p(1-p)}{N}} \approx \sqrt{\frac{\hat{T}(1-\hat{T})}{N}} \tag{21}$$

This is identical in structure to the uncertainty expression derived in Stage I for π estimation. The $1/\sqrt{N}$ convergence is not specific to either problem but a universal consequence of the Central Limit Theorem: averaging N independent random variables always produces uncertainty scaling as $1/\sqrt{N}$, regardless of the underlying physical model.

4 Simulation Methodology

4.1 Implementation

The simulation was implemented in Python using NumPy for vectorised array operations. For each neutron history, a uniform random number r is drawn and converted to a free path length via Equation (17). The neutron is counted as transmitted if $d > x$, and absorbed otherwise. The logical structure is as follows:

```
function simulate(N, x, Sigma):
    r           <- draw N uniform random numbers in (0, 1)
    free_paths  <- -log(r) / Sigma
    transmitted <- count(free_paths > x)
    T_mc        <- transmitted / N
    sigma_T     <- sqrt(T_mc * (1 - T_mc) / N)
    return T_mc, sigma_T
```

NumPy vectorisation applies this procedure to all N neutrons simultaneously, giving $O(N)$ time complexity.

4.2 Experimental Design

Three experiments were conducted, as described in Table 2.

Experiment	**Description**
Thickness sweep	$N = 500{,}000$, $\Sigma = 1.0$ cm^{-1}, $x \in [0, 5]$ cm (60 points). Validates $P(x) = e^{-\Sigma x}$ across the full thickness range.
Convergence study	$x = 1.0$ cm, $\Sigma = 1.0$ cm^{-1}, $N \in [10^2, 10^6]$ (9 points, log-spaced). Measures σ_T scaling with N.
Material comparison	$N = 300{,}000$, $x \in [0, 10]$ cm, four materials from Table 1. Demonstrates the dependence of attenuation on Σ.

Table 2: Experimental parameters for each simulation study.

5 Results

5.1 Transmission vs Thickness

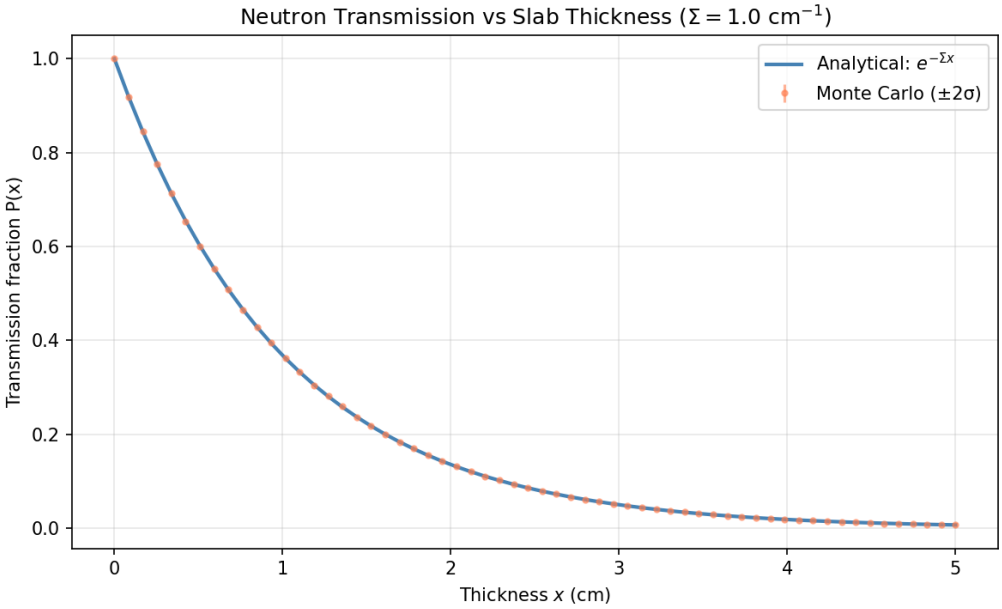

Figure 1: Monte Carlo transmission fraction vs slab thickness for $\Sigma = 1.0$ cm^{-1}, $N = 500{,}000$. Orange points: MC estimates with $\pm 2\sigma_T$ error bars. Blue curve: analytical result $P(x) = e^{-\Sigma x}$.

The results show a clear exponential decay of transmission with increasing slab thickness, consistent with $P(x) = e^{-\Sigma x}$. As shown in Figure 1, the Monte Carlo data points lie directly on the analytical curve across the full range. At $x = 1$ cm (one mean free path), the analytical transmission is $e^{-1} \approx 0.368$; at $x = 2$ cm it is $e^{-2} \approx 0.135$; and at $x = 5$ cm it falls to $e^{-5} \approx 0.007$. All MC estimates fall within $\pm 2\sigma_T$ of the analytical value, consistent with the expected 95% coverage of a two-sigma interval. The error bars are smallest at large x, where $\hat{T}$ is near zero and the Bernoulli variance $\hat{T}(1-\hat{T})$ is minimised.

5.2 Statistical Convergence

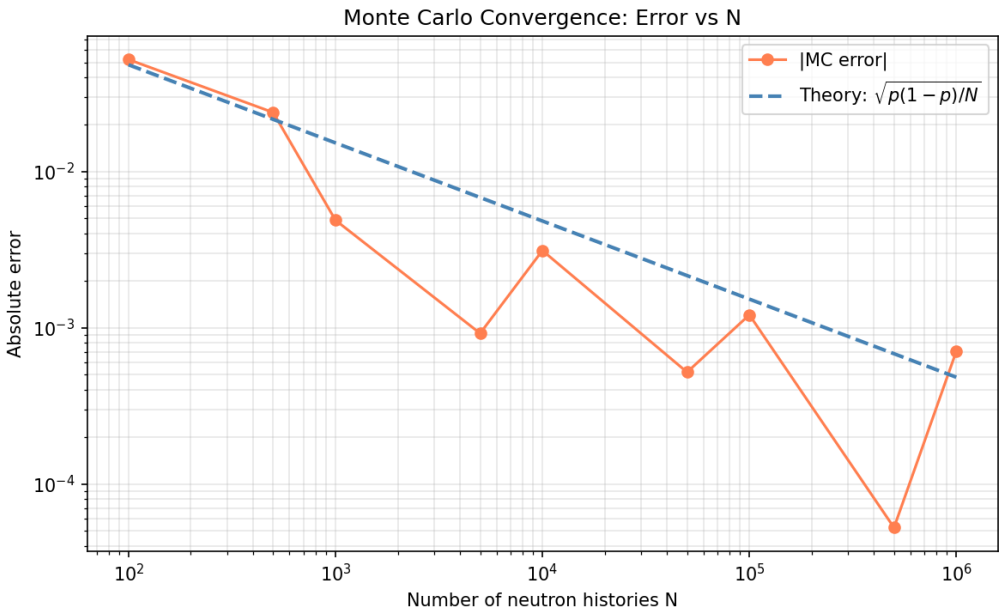

Figure 2: Absolute MC error vs number of neutron histories N on log-log axes. Orange: observed error $|\hat{T} - e^{-\Sigma x}|$. Dashed blue: theoretical uncertainty $\sqrt{p(1-p)/N}$.

Figure 2 shows the decrease in MC error with increasing N, consistent with the $1/\sqrt{N}$ relationship. The error curve is irregular rather than smooth, as each data point represents a single random realisation. At any given N, the actual error is itself a random variable and may fall above or below the theoretical envelope. The overall slope across the full range is approximately -0.5 on log-log axes, confirming $\sigma_T \propto 1/\sqrt{N}$.

This behaviour is identical to the convergence observed in Stage I π estimation. The $1/\sqrt{N}$ scaling is not a feature of the neutron transport model but a universal property of Monte Carlo estimation, as described in Section 3

N	$\hat{T}$	$T_{\text{analytical}}$	σ_T
10^2	0.37000	0.36788	0.04828
10^3	0.37400	0.36788	0.01530
10^4	0.36770	0.36788	0.00482
10^5	0.36979	0.36788	0.00153
10^6	0.36759	0.36788	0.00048

Table 3: Monte Carlo transmission estimates at selected sample sizes for $\Sigma = 1.0$ cm^{-1}, $x = 1.0$ cm. All estimates fall within σ_T of the analytical value $e^{-1} = 0.36788$.

5.3 Material Comparison

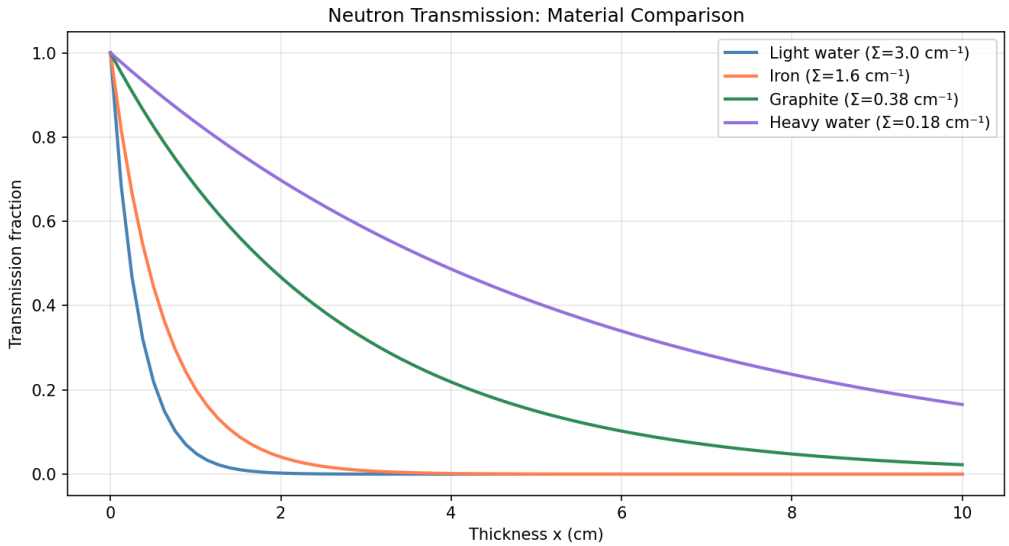

Figure 3: Neutron transmission vs slab thickness for four reactor-relevant materials. Steeper decay corresponds to larger Σ.

Figure 3 illustrates the strong dependence of attenuation rate on Σ. Light water ($\Sigma = 3.0$ cm^{-1}) reduces transmission to below 5% within 1 cm; heavy water ($\Sigma = 0.18$ cm^{-1}) retains 17% transmission at 10 cm. Table 4 lists the thickness required for each material to reach 10% transmission, computed from $x_{10\%} = \ln(10)/\Sigma$.

Material	Σ (cm^{-1})	$x_{10\%} = \ln(10)/\Sigma$ (cm)
Light water	3.00	0.77
Iron	1.60	1.44
Graphite	0.38	6.06
Heavy water	0.18	12.79

Table 4: Slab thickness required to reduce neutron transmission to 10% for each material.

6 Discussion

The results confirm that the Monte Carlo simulation correctly recovers exponential attenuation across all tested geometries and materials. The agreement between $\hat{T}$ and $e^{-\Sigma x}$ to within σ_T in every experiment demonstrates that the physics is correctly captured by the single-collision model.

The most significant observation is that the exponential attenuation law was not assumed at any stage. It was derived from one physical postulate — the memoryless, constant collision rate — and independently recovered through random sampling. The

simulation therefore constitutes a numerical verification of the derivation in Section 2.2, not merely a calculation of a known result. It is worth noting that the same mathematical structure governs light attenuation in absorbing media, described by the identical Beer-Lambert law from classical optics. The underlying stochastic process is the same: a memoryless absorption rate per unit path length.

The $1/\sqrt{N}$ convergence observed in Figure 2 is identical to the convergence found in Stage I π estimation. The two problems are physically unrelated, yet their Monte Carlo estimators share the same statistical behaviour. This is a direct consequence of the Central Limit Theorem and reflects the universal structure common to all Monte Carlo methods: the estimator is always a mean of independent random variables, and its uncertainty always scales as $1/\sqrt{N}$.

The material comparison reveals the practical importance of Σ in reactor design. The ratio of the heavy water to light water mean free paths is $5.56/0.33 \approx 17$. This difference explains why CANDU reactors, which use heavy water as a moderator, can sustain a chain reaction with natural (unenriched) uranium, while light-water reactors require fuel enriched to 3–5% ^{235}U. Light water's high Σ means it absorbs too many neutrons for natural uranium to sustain criticality.

The primary limitation of this model is that every collision is treated as absorption. In reality, neutrons may scatter, changing direction and energy and continuing transport. In materials with high scattering-to-absorption ratios, such as graphite and heavy water, scattered neutrons contribute significantly to total transmission. The current model computes the uncollided flux only. This limitation motivates Stage III, in which full multi-collision transport with scattering, fission, and energy-dependent cross-sections will be implemented using OpenMC.

7 Computational Analysis

As with Stage I, execution time scales approximately linearly with N:

$$\text{Time} \propto N \tag{22}$$

This is a consequence of the $O(N)$ vectorised implementation, in which all N free paths are sampled and compared simultaneously. The accuracy-cost tradeoff identified in Stage I therefore persists in Stage II: halving the uncertainty requires a fourfold increase in N, and consequently a fourfold increase in computation time.

8 Progression to Stage III: Full Reactor Simulation

The Stage II model is the conceptual skeleton of the transport kernel used in production Monte Carlo codes. OpenMC extends it in three directions: first, scattering events redirect neutrons with a new energy and direction sampled from nuclear scattering distributions, rather than terminating their history; second, the cross-section Σ becomes a function of neutron energy, requiring continuous-energy evaluated nuclear data (ENDF/B-VIII.0); and third, the slab geometry is replaced by arbitrary three-dimensional geometry defined by constructive solid geometry.

Stage III will simulate a uranium fuel pincell — a cylindrical UO_2 fuel rod surrounded by a water moderator — and compute the criticality eigenvalue k_{eff}: the ratio of neutrons produced in one fission generation to those in the previous generation. The condition $k_{\text{eff}} = 1$ describes a sustained, self-regulating chain reaction and is the fundamental design criterion for every nuclear reactor in operation.

9 Conclusion

A Monte Carlo neutron transport simulation was developed from first principles and validated against the analytical Beer-Lambert attenuation law. Results demonstrate agreement between Monte Carlo estimates and analytical predictions to within the theoretical statistical uncertainty across all tested geometries and materials, confirming the physical correctness of the simulation.

The $1/\sqrt{N}$ convergence identified in Stage I Monte Carlo π estimation reappears identically in Stage II neutron transport, confirming that this convergence rate is a universal property of Monte Carlo estimation governed by the Central Limit Theorem and independent of the underlying physical model. This observation forms the central mathematical thread connecting the two stages of this research project.

This single-collision transport model establishes the conceptual and computational foundation for Stage III full reactor simulation in OpenMC, in which scattering, energy dependence, and three-dimensional geometry will be incorporated to compute reactor criticality from first principles.

References

[1] E.E. Lewis, *Fundamentals of Nuclear Reactor Physics*. Academic Press, 2008.

[2] J.R. Lamarsh and A.J. Baratta, *Introduction to Nuclear Engineering*, 3rd ed. Prentice Hall, 2001.

[3] P.K. Romano et al., "OpenMC: A State-of-the-Art Monte Carlo Code for Research and Development," *Annals of Nuclear Energy*, vol. 82, pp. 90–97, 2015.

[4] N. Metropolis and S. Ulam, "The Monte Carlo Method," *Journal of the American Statistical Association*, vol. 44, no. 247, pp. 335–341, 1949.

[5] International Atomic Energy Agency, *Neutron Cross Section Libraries*, IAEA-NDS. Vienna, 2023.